```
> # DISCOVERY OF IDENTITIES FOR a_7(n)
  with(qseries) :
  T := 60 : N := 7 * T + 20 :
  f := d → etaq(q, d, N) :
  tq := q * f(7)^4 / f(1)^4 :
  p0q := q * f(14)^4 * f(1)^4 / (f(7)^4 * f(2)^4) :
  p1q := (f(14) * f(1)^7 / (f(7) * f(2)^7) − 1) / 7 :
  A7q := f(2)^6 * f(49)^7 / (q^10 * f(1)^7 * f(98)^6) :
  B7q := 1 / q :

> # GROUP 1, IDENTITY 1: U_A7(p0)
  m := 6 :
  S7_1_1 := series(A7q * p0q, q, 7 * T) :
  target7_1_1 := series(tq * (add(coeff(S7_1_1, q, 7 * n) * q^n, n = −1 .. T − 1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols7 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R7_1_1 := expand(findlincombo(target7_1_1, basis7, symbols7, q, 0) / t) :
  collect(R7_1_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&\left((7)^8 t^5 + (2)\,(7)^6\,(17)\,t^4 + (3)^4\,(5)\,(7)^4\,t^3 + (2)\,(5)^2\,(7)^2\,(43)\,t^2 + (2)^4\,(7)\,(43)\,t\right.\\&\left. + (2)^3\,(7)\right)p0 + \left(-(7)^7 t^4 - (3)\,(7)^5\,(11)\,t^3 - (3)\,(7)^3\,(127)\,t^2 - (2)^3\,(5)\,(7)^3\,t\right.\\&\left. - (3)\,(7)\,(29) - \frac{(7)}{t}\right)p1 + (3)^2\,(7)^6\,t^4 + (3)\,(5)\,(7)^4\,(17)\,t^3 + (3)\,(7)^3\,(131)\,t^2\\&+ (2)^4\,(3)\,(7)\,(41)\,t + (3)\,(7)\,(29) + \frac{(7)}{t}\end{aligned} \qquad (1)$$

```
> # GROUP 1, IDENTITY 2: U_A7(p0*t^(-1))
  m := 5 :
  S7_1_2 := series(A7q * p0q / tq, q, 7 * T) :
  target7_1_2 := series(tq * (add(coeff(S7_1_2, q, 7 * n) * q^n, n = −1 .. T − 1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols7 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R7_1_2 := expand(findlincombo(target7_1_2, basis7, symbols7, q, 0) / t) :
  collect(R7_1_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((7)^3 t^2 + (2)^3 (3) (7) t + (2)^4\right) p0 + \left(-(7)^2 t - (23) - \frac{(2)}{t}\right) p1 + (3)^2 (7) t + (23) \quad (2)$$
$$+ \frac{(2)}{t}$$

```
> # GROUP 1, IDENTITY 3: U_A7(p0*t^(-2))
  m := 5 :
  S7_1_3 := series(A7q * p0q / tq^2, q, 7 * T) :
  target7_1_3 := series(tq * (add(coeff(S7_1_3, q, 7 * n) * q^n, n = -1 .. T - 1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols7 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R7_1_3 := expand(findlincombo(target7_1_3, basis7, symbols7, q, 0) / t) :
  collect(R7_1_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$\left(-(7)^5 t^3 - (2)^2 (7)^3 (11) t^2 - (2)^4 (7) (29) t - (2)^4 (13)\right) p0 + \left((7)^4 t^2 + (7)^2 (43) t \quad (3)\right.$$
$$\left. + (2)^4 (3)^3 + \frac{(2) (13)}{t}\right) p1 - (7)^3 (11) t^2 - (7)^2 (47) t - (2)^4 (3)^3 - \frac{(2) (13)}{t}$$

```
> # GROUP 1, IDENTITY 4: U_A7(p0*t^(-3))
  m := 5 :
  S7_1_4 := series(A7q * p0q / tq^3, q, 7 * T) :
  target7_1_4 := series(tq * (add(coeff(S7_1_4, q, 7 * n) * q^n, n = -1 .. T - 1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q
      * tq^j, q, T), j = 0 .. m)] :
  symbols7 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
  R7_1_4 := expand(findlincombo(target7_1_4, basis7, symbols7, q, 0) / t) :
  collect(R7_1_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$\left(-(7)^7 t^4 - (2)^5 (7)^5 t^3 - (2) (3)^3 (7)^3 t^2 + (2)^4 (7) (131) t + (2)^6 (3) (7)\right) p0 + \left((7)^6 t^3 \quad (4)\right.$$
$$\left. + (7)^4 (31) t^2 + (2) (7)^2 (17) t - (2) (7) (11) (13) - \frac{(2)^3 (3) (7)}{t}\right) p1$$
$$- (7)^5 (11) t^3 - (7)^3 (11) (19) t^2 - (2)^3 (7)^2 t + (2) (7) (11) (13) + \frac{(2)^3 (3) (7)}{t}$$

```
> # GROUP 1, IDENTITY 5: U_A7(p0*t^(-4))
```

```
m := 5 :
S7_1_5 := series(A7q * p0q / tq^4, q, 7 * T) :
target7_1_5 := series(tq * (add(coeff(S7_1_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
    T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R7_1_5 := expand( findlincombo(target7_1_5, basis7, symbols7, q, 0) / t) :
collect(R7_1_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\begin{aligned}&\left((2)\,(7)^7\,(13)\,t^4 + (3)\,(7)^5\,(13)\,(19)\,t^3 + (2)\,(3)\,(7)^3\,(839)\,t^2 + (7)^2\,(353)\,t\right.\\&\left. - (2)^5\,(3)\,(67)\right)p0 + \left(-(2)\,(7)^6\,(13)\,t^3 - (3)\,(7)^4\,(239)\,t^2 - (2)^2\,(7)^3\,(163)\,t\right.\\&\left. - (2)\,(7)\,(97) + \frac{(2)^2\,(3)\,(67)}{t}\right)p1 + (2)\,(7)^7\,t^4 + (2)\,(7)^5\,(131)\,t^3 + (7)^5\,(107)\,t^2\\&+ (2)^3\,(3)\,(5)\,(7)^2\,(37)\,t + (2)\,(7)\,(97) - \frac{(2)^2\,(3)\,(67)}{t}\end{aligned} \tag{5}$$

```
> # GROUP 1, IDENTITY 6: U_A7(p0*t^(-5))
m := 8 :
S7_1_6 := series(A7q * p0q / tq^5, q, 7 * T) :
target7_1_6 := series(tq^2 * (add(coeff(S7_1_6, q, 7 * n) * q^n, n = -2 ..T-1) + O(q^T) ), q,
    T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R7_1_6 := expand( findlincombo(target7_1_6, basis7, symbols7, q, 0) / t^2) :
collect(R7_1_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

$$\begin{aligned}&\left((7)^{11}\,t^6 + (2)\,(7)^9\,(13)\,t^5 - (2)\,(7)^7\,(47)\,t^4 - (2)\,(3)^2\,(5)\,(7)^6\,(13)\,t^3\right.\\&\left. - (2)\,(7)^4\,(5011)\,t^2 - (2)\,(7)^2\,(73)\,(157)\,t + (2)^3\,(3)\,(673) + \frac{(2)^3}{t}\right)p0 + \left(\right.\\&\left. -(7)^{10}\,t^5 - (5)^2\,(7)^8\,t^4 + (2)^4\,(7)^7\,t^3 + (7)^6\,(163)\,t^2 + (2)^2\,(5)\,(7)^3\,(461)\,t\right.\\&\left. + (7)\,(53)\,(373) - \frac{(2)^2\,(5)\,(101)}{t} - \frac{1}{t^2}\right)p1 + (7)^{10}\,t^5 + (7)^7\,(139)\,t^4\\&- (2)^2\,(3)^2\,(7)^7\,t^3 - (7)^4\,(8539)\,t^2 - (2)^4\,(5)\,(7)^2\,(11)\,(73)\,t - (3)^3\,(5)^3\,(41)\end{aligned} \tag{6}$$

$$+\frac{(2)^2\,(5)\,(101)}{t}+\frac{1}{t^2}$$

```
> # GROUP 1, IDENTITY 7: U_A7(p0*t^(-6))
  m := 9 :
  S7_1_7 := series(A7q * p0q / tq^6, q, 7 * T) :
  target7_1_7 := series(tq^2 * (add(coeff(S7_1_7, q, 7 * n) * q^n, n = -2 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_1_7 := expand( findlincombo(target7_1_7, basis7, symbols7, q, 0) / t^2) :
  collect(R7_1_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 30$$

$$\#\ of\ terms\ , 51$$

$$\left((7)^{13}\,t^7+(2)^5\,(7)^{11}\,t^6+(5)\,(7)^9\,(89)\,t^5+(2)\,(7)^7\,(3671)\,t^4+(3)\,(5)\,(7)^6\,(29)\,(37)\,t^3\right. \tag{7}$$

$$+(2)\,(7)^4\,(37)\,(1567)\,t^2+(3)\,(7)^2\,(13)\,(41)\,(197)\,t+(2)^3\,(3)^2\,(1523)$$

$$\left.-\frac{(2)^3\,(3)\,(7)}{t}\right)p0+\left(-(7)^{12}\,t^6-(7)^{10}\,(31)\,t^5-(5)^2\,(7)^8\,(17)\,t^4-(7)^7\,(1019)\,t^3\right.$$

$$-(2)^2\,(3)\,(5)\,(7)^5\,(257)\,t^2-(2)\,(7)^3\,(53407)\,t-(7)\,(277603)-\frac{(2)\,(3)\,(2281)}{t}$$

$$\left.+\frac{(3)\,(7)}{t^2}\right)p1+(7)^{11}\,(11)\,t^6+(3)^3\,(7)^9\,(11)\,t^5+(7)^8\,(571)\,t^4+(3)\,(7)^6\,(2927)\,t^3$$

$$+(2)^2\,(7)^4\,(17)\,(41)^2\,t^2+(2)\,(7)^2\,(29)\,(67)\,(193)\,t+(7)\,(277579)$$

$$+\frac{(2)\,(3)\,(2281)}{t}-\frac{(3)\,(7)}{t^2}$$

```
> # GROUP 2, IDENTITY 1: U_A7(p1)
  m := 9 :
  S7_2_1 := series(A7q * p1q, q, 7 * T) :
  target7_2_1 := series(tq * (add(coeff(S7_2_1, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_2_1 := expand( findlincombo(target7_2_1, basis7, symbols7, q, 0) / t) :
  collect(R7_2_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 30$$

$$\#\ of\ terms\ , 51$$

$$\left(-(2)\,(3)\,(7)^{13}\,t^8-(2)^2\,(7)^{11}\,(53)\,t^7-(2)\,(3)\,(7)^9\,(13)\,(41)\,t^6-(2)^2\,(7)^8\,(947)\,t^5\right. \tag{8}$$
$$-(2)\,(3)^2\,(7)^6\,(13)\,(79)\,t^4-(2)^5\,(3)\,(7)^4\,(547)\,t^3-(2)^2\,(3)\,(7)^2\,(13)\,(509)\,t^2$$
$$\left.-(2)^4\,(3)\,(7)\,(149)\,t-(2)^4\,(7)\right)p0+\left((2)\,(3)\,(7)^{12}\,t^7+(3)^2\,(7)^{10}\,(23)\,t^6\right.$$
$$+(7)^8\,(43)\,(71)\,t^5+(2)^2\,(3)\,(5)\,(7)^7\,(59)\,t^4+(3)\,(5)\,(7)^5\,(1129)\,t^3$$
$$\left.+(5)\,(7)^3\,(11)\,(859)\,t^2+(3)\,(7)\,(47)\,(499)\,t+(2)^7\,(7)^2+\frac{(2)\,(7)}{t}\right)p1-(7)^{13}\,t^8$$
$$-(2)\,(5)\,(7)^{12}\,t^7-(3)^2\,(7)^9\,(197)\,t^6-(3)\,(7)^8\,(11)\,(101)\,t^5-(3)\,(7)^6\,(13)\,(659)\,t^4$$
$$-(3)\,(7)^4\,(13)\,(3079)\,t^3-(5)\,(7)^2\,(66347)\,t^2-(5)^3\,(7)\,(563)\,t-(2)^7\,(7)^2$$
$$-\frac{(2)\,(7)}{t}$$

```
> # GROUP 2, IDENTITY 2: U_A7(p1*t^(-1))
  m := 5 :
  S7_2_2 := series(A7q * p1q / tq, q, 7 * T) :
  target7_2_2 := series(tq * (add(coeff(S7_2_2, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_2_2 := expand( findlincombo(target7_2_2, basis7, symbols7, q, 0) / t) :
  collect(R7_2_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)\,t-(2)^3\right)p0+\left((2)^3+\frac{1}{t}\right)p1-(7)\,t-(2)^3-\frac{1}{t} \tag{9}$$

```
> # GROUP 2, IDENTITY 3: U_A7(p1*t^(-2))
  m := 5 :
  S7_2_3 := series(A7q * p1q / tq^2, q, 7 * T) :
  target7_2_3 := series(tq * (add(coeff(S7_2_3, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_2_3 := expand( findlincombo(target7_2_3, basis7, symbols7, q, 0) / t) :
  collect(R7_2_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

(10)

$$\left((2)^3\,(7)^3\,t^2+(2)^3\,(3)\,(5)\,(7)\,t+(2)^6\right)p0+\left(-(2)^3\,(7)^2\,t-(113)-\frac{(2)^3}{t}\right)p1 \qquad (10)$$
$$+(7)^3\,t^2+(2)^6\,(7)\,t+(113)+\frac{(2)^3}{t}$$

```
> # GROUP 2, IDENTITY 4: U_A7(p1*t^(-3))
  m := 5 :
  S7_2_4 := series(A7q * p1q / tq^3, q, 7 * T) :
  target7_2_4 := series(tq * (add(coeff(S7_2_4, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_2_4 := expand( findlincombo(target7_2_4, basis7, symbols7, q, 0) / t) :
  collect(R7_2_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$\left((2)^2\,(7)^5\,t^3-(2)^4\,(3)\,(7)^3\,t^2-(2)^5\,(5)\,(7)^2\,t-(2)^4\,(3)\,(13)\right)p0+\left(-(2)^2\,(7)^4\,t^2 \qquad (11)$$
$$+(3)\,(7)^2\,(17)\,t+(2)\,(23)^2+\frac{(2)\,(3)\,(13)}{t}\right)p1+(7)^5\,t^3+(2)^2\,(5)\,(7)^3\,t^2$$
$$-(7)\,(439)\,t-(2)\,(23)^2-\frac{(2)\,(3)\,(13)}{t}$$

```
> # GROUP 2, IDENTITY 5: U_A7(p1*t^(-4))
  m := 5 :
  S7_2_5 := series(A7q * p1q / tq^4, q, 7 * T) :
  target7_2_5 := series(tq * (add(coeff(S7_2_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_2_5 := expand( findlincombo(target7_2_5, basis7, symbols7, q, 0) / t) :
  collect(R7_2_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^7\,t^4-(2)^3\,(7)^5\,(31)\,t^3-(2)^3\,(7)^3\,(191)\,t^2+(2)^5\,(3)^2\,(7)^2\,t \qquad (12)$$
$$+(2)^5\,(7)\,(17)\right)p0+\left((2)^3\,(7)^6\,t^3+(7)^4\,(241)\,t^2+(2)^3\,(3)^2\,(7)^2\,(19)\,t\right.$$
$$\left.-(2)^6\,(5)\,(7)-\frac{(2)^2\,(7)\,(17)}{t}\right)p1-(7)^7\,t^4-(2)^3\,(7)^5\,(11)\,t^3-(3)\,(7)^4\,(83)\,t^2$$

$$-(2)^4\,(3)^4\,(7)^2\,t+(2)^6\,(5)\,(7)+\frac{(2)^2\,(7)\,(17)}{t}$$

```
> # GROUP 2, IDENTITY 6: U_A7(p1*t^(-5))
  m := 7 :
  S7_2_6 := series(A7q*p1q/tq^5, q, 7*T) :
  target7_2_6 := series(tq^2*(add(coeff(S7_2_6, q, 7*n)*q^n, n=-2..T-1)+O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
      *tq^j, q, T), j=0..m)] :
  symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
  R7_2_6 := expand(findlincombo(target7_2_6, basis7, symbols7, q, 0)/t^2) :
  collect(R7_2_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\left((2)\,(3)\,(7)^9\,t^5+(2)^2\,(7)^7\,(73)\,t^4+(2)\,(3)\,(7)^6\,(127)\,t^3+(2)^2\,(7)^4\,(1279)\,t^2\right. \tag{13}$$

$$+(2)\,(7)^2\,(59)\,(79)\,t-(2)^3\,(3)\,(613)-\left.\frac{(2)^3}{t}\right)p0+\left(-(2)\,(3)\,(7)^8\,t^4\right.$$

$$-(7)^7\,(41)\,t^3-(7)^5\,(17)\,(43)\,t^2-(2)^3\,(3)^2\,(5)\,(7)^3\,(13)\,t-(5)\,(7)\,(1579)$$

$$\left.+\frac{(2)^4\,(5)\,(23)}{t}+\frac{1}{t^2}\right)p1+(7)^9\,t^5+(2)\,(3)\,(7)^7\,(13)\,t^4+(7)^6\,(367)\,t^3$$

$$+(3)^3\,(7)^4\,(199)\,t^2+(2)^3\,(7)^2\,(31)\,(131)\,t+(3)\,(113)\,(163)-\frac{(2)^4\,(5)\,(23)}{t}$$

$$-\frac{1}{t^2}$$

```
> # GROUP 2, IDENTITY 7: U_A7(p1*t^(-6))
  m := 8 :
  S7_2_7 := series(A7q*p1q/tq^6, q, 7*T) :
  target7_2_7 := series(tq^2*(add(coeff(S7_2_7, q, 7*n)*q^n, n=-2..T-1)+O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
      *tq^j, q, T), j=0..m)] :
  symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
  R7_2_7 := expand(findlincombo(target7_2_7, basis7, symbols7, q, 0)/t^2) :
  collect(R7_2_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

(14)

$$\left(-(2)^2\,(7)^{11}\,t^6-(2)^3\,(7)^9\,(29)\,t^5-(2)^2\,(7)^7\,(1481)\,t^4-(2)^4\,(3)\,(5)\,(7)^6\,(47)\,t^3\right. \tag{14}$$

$$-(2)^2\,(7)^4\,(17551)\,t^2-(2)^3\,(7)^2\,(11)\,(31)\,(61)\,t-(2)^4\,(3)^3\,(29)+\frac{(2)^5\,(5)}{t}\Bigg)\,p0$$

$$+\left((2)^2\,(7)^{10}\,t^5+(7)^8\,(229)\,t^4+(2)\,(3)\,(7)^7\,(137)\,t^3+(2)^2\,(7)^6\,(383)\,t^2\right.$$

$$\left.+(2)\,(5)\,(7)^3\,(41)\,(157)\,t+(2)\,(5)^2\,(7)\,(2917)+\frac{(2)\,(773)}{t}-\frac{(2)^2\,(5)}{t^2}\right)\,p1$$

$$-(7)^{11}\,t^6-(2)^2\,(7)^9\,(17)\,t^5-(7)^7\,(23)\,(101)\,t^4-(2)\,(3)\,(5)\,(7)^6\,(227)\,t^3$$

$$-(2)\,(3)\,(7)^4\,(13103)\,t^2-(2)\,(7)^2\,(225479)\,t-(2)\,(5)\,(102079)-\frac{(2)\,(773)}{t}$$

$$+\frac{(2)^2\,(5)}{t^2}$$

```
> # GROUP 3, IDENTITY 1: U_A7(1)
  m := 9 :
  S7_3_1 := series(A7q, q, 7 * T) :
  target7_3_1 := series(tq * (add(coeff(S7_3_1, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_3_1 := expand( findlincombo(target7_3_1, basis7, symbols7, q, 0) / t) :
  collect(R7_3_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 30$$

$$\#\ of\ terms\ , 51$$

$$\left((2)\,(3)\,(7)^{14}\,t^8+(2)^2\,(7)^{12}\,(53)\,t^7+(2)\,(3)\,(7)^{10}\,(13)\,(41)\,t^6+(2)^2\,(7)^9\,(947)\,t^5\right. \tag{15}$$

$$+(2)\,(3)^2\,(7)^7\,(13)\,(79)\,t^4+(2)^5\,(3)\,(7)^5\,(547)\,t^3+(2)^2\,(3)\,(7)^3\,(13)\,(509)\,t^2$$

$$\left.+(2)^3\,(7)\,(11)\,(569)\,t+(2)^4\,(7)^2\right)\,p0+\left(-(2)\,(3)\,(7)^{13}\,t^7-(3)^2\,(7)^{11}\,(23)\,t^6\right.$$

$$-(7)^9\,(43)\,(71)\,t^5-(2)^2\,(3)\,(5)\,(7)^8\,(59)\,t^4-(3)\,(5)\,(7)^6\,(1129)\,t^3$$

$$-(2)\,(3)^2\,(7)^3\,(19)\,(967)\,t^2-(3)\,(7)^2\,(47)\,(499)\,t-(3)^2\,(7)\,(17)\,(41)$$

$$\left.-\frac{(2)\,(7)^2}{t}\right)\,p1+(7)^{14}\,t^8+(2)\,(5)\,(7)^{13}\,t^7+(3)^2\,(7)^{10}\,(197)\,t^6$$

$$+(3)\,(7)^9\,(11)\,(101)\,t^5+(3)\,(7)^7\,(13)\,(659)\,t^4+(3)\,(7)^5\,(13)\,(3079)\,t^3$$

$$+(2)\,(7)^2\,(29)\,(40037)\,t^2+(5)^3\,(7)^2\,(563)\,t+(3)^2\,(7)\,(17)\,(41)+\frac{(2)\,(7)^2}{t}$$

```
> # GROUP 3, IDENTITY 2: U_A7(t^(-1))
  m := 5 :
  S7_3_2 := series(A7q/tq, q, 7*T) :
  target7_3_2 := series(tq*(add(coeff(S7_3_2, q, 7*n)*q^n, n=-1..T-1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
      *tq^j, q, T), j=0..m)] :
  symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
  R7_3_2 := expand(findlincombo(target7_3_2, basis7, symbols7, q, 0)/t) :
  collect(R7_3_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((2)^4\ (3)\ (7)\ t + (2)^4\ (3)\right) p0 + \left(-(2)^4\ (3) - \frac{(2)\ (3)}{t}\right) p1 + (7)^2\ t + (2)^4\ (3) \qquad \textbf{(16)}$$

$$+ \frac{(2)\ (3)}{t}$$

```
> # GROUP 3, IDENTITY 3: U_A7(t^(-2))
  m := 5 :
  S7_3_3 := series(A7q/tq^2, q, 7*T) :
  target7_3_3 := series(tq*(add(coeff(S7_3_3, q, 7*n)*q^n, n=-1..T-1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
      *tq^j, q, T), j=0..m)] :
  symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
  R7_3_3 := expand(findlincombo(target7_3_3, basis7, symbols7, q, 0)/t) :
  collect(R7_3_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(2)^4\ (5)\ (7)\ t - (2)^6\right) p0 + \left((2)\ (3)\ (13) + \frac{(2)^3}{t}\right) p1 + (7)^3\ t^2 - (2)^3\ (7)\ t \qquad \textbf{(17)}$$

$$- (2)\ (3)\ (13) - \frac{(2)^3}{t}$$

```
> # GROUP 3, IDENTITY 4: U_A7(t^(-3))
  m := 5 :
  S7_3_4 := series(A7q/tq^3, q, 7*T) :
  target7_3_4 := series(tq*(add(coeff(S7_3_4, q, 7*n)*q^n, n=-1..T-1) + O(q^T)), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
      *tq^j, q, T), j=0..m)] :
  symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)] :
```

```
R7_3_4 := expand( findlincombo(target7_3_4, basis7, symbols7, q, 0) / t) :
collect(R7_3_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^2\,(3)\,(7)^5\,t^3 + (2)^3\,(3)\,(7)^3\,(13)\,t^2 + (2)^4\,(5)^2\,(7)^2\,t + (2)^5\,(5)\,(7)\right) p0 + \left(\tag{18}$$

$$-(2)^2\,(3)\,(7)^4\,t^2 - (2)\,(7)^2\,(151)\,t - (2)\,(5)\,(7)\,(37) - \frac{(2)^2\,(5)\,(7)}{t}\right) p1 + (7)^5\,t^3$$

$$+ (2)^2\,(3)^3\,(7)^3\,t^2 + (2)^3\,(7)\,(283)\,t + (2)\,(5)\,(7)\,(37) + \frac{(2)^2\,(5)\,(7)}{t}$$

```
> # GROUP 3, IDENTITY 5: U_A7(t^(-4))
  m := 6 :
  S7_3_5 := series(A7q / tq^4, q, 7 * T) :
  target7_3_5 := series(tq^2 * (add(coeff(S7_3_5, q, 7 * n) * q^n, n = -2 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_3_5 := expand( findlincombo(target7_3_5, basis7, symbols7, q, 0) / t^2) :
  collect(R7_3_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left(-(2)^3\,(3)^3\,(7)^4\,t^2 - (2)^3\,(7)^2\,(347)\,t - (2)^3\,(5)\,(7)\,(31) + \frac{(2)^3}{t}\right) p0 + \left((7)^4\,t^2 \tag{19}$$

$$+ (2)^3\,(3)^3\,(7)^3\,t + (5)\,(7)\,(11)\,(47) + \frac{(2)^2\,(271)}{t} - \frac{1}{t^2}\right) p1 - (7)^7\,t^4$$

$$-(7)^3\,(167)\,t^2 - (2)^4\,(3)\,(5)\,(7)^3\,t - (3)\,(6029) - \frac{(2)^2\,(271)}{t} + \frac{1}{t^2}$$

```
> # GROUP 3, IDENTITY 6: U_A7(t^(-5))
  m := 7 :
  S7_3_6 := series(A7q / tq^5, q, 7 * T) :
  target7_3_6 := series(tq^2 * (add(coeff(S7_3_6, q, 7 * n) * q^n, n = -2 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_3_6 := expand( findlincombo(target7_3_6, basis7, symbols7, q, 0) / t^2) :
  collect(R7_3_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

of terms , 45

$$\left((2)\,(3)\,(7)^9\,t^5+(2)^2\,(7)^7\,(19)\,t^4-(2)\,(3)^2\,(7)^7\,t^3-(2)^2\,(5)\,(7)^4\,(29)\,t^2\right. \tag{20}$$

$$\left.+(2)\,(7)^2\,(5449)\,t+(2)^3\,(3)\,(2161)-\frac{(2)^3\,(13)}{t}\right)p0+\left(-(2)\,(3)\,(7)^8\,t^4\right.$$

$$\left.-(2)\,(5)\,(7)^7\,t^3+(3)\,(7)^5\,(43)\,t^2+(2)\,(7)^3\,(223)\,t-(5)^2\,(7)\,(419)\right.$$

$$\left.-\frac{(2)\,(5)\,(647)}{t}+\frac{(13)}{t^2}\right)p1+(7)^9\,t^5+(2)\,(7)^7\,(31)\,t^4+(2)\,(3)^3\,(7)^6\,t^3$$

$$-(7)^4\,(13)\,(67)\,t^2-(2)\,(7)^3\,(151)\,t+(3)\,(24407)+\frac{(2)\,(5)\,(647)}{t}-\frac{(13)}{t^2}$$

```
> # GROUP 3, IDENTITY 7: U_A7(t^(-6))
  m := 8 :
  S7_3_7 := series(A7q / tq^6, q, 7 * T) :
  target7_3_7 := series(tq^2 * (add(coeff(S7_3_7, q, 7 * n) * q^n, n = -2 ..T-1) + O(q^T) ), q,
      T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R7_3_7 := expand( findlincombo(target7_3_7, basis7, symbols7, q, 0) / t^2 ) :
  collect(R7_3_7, [p0, p1, t], recursive, ifactor);
```

nx = , 27

of terms , 48

$$\left(-(2)^2\,(3)\,(7)^{11}\,t^6-(2)^3\,(5)\,(7)^9\,(11)\,t^5-(2)^2\,(5)^2\,(7)^9\,t^4-(2)^3\,(7)^7\,(13)\,t^3\right. \tag{21}$$

$$\left.+(2)^2\,(3)\,(7)^4\,(1451)\,t^2+(2)^3\,(3)\,(7)^2\,(463)\,t-(2)^3\,(7)\,(37)\,(113)\right.$$

$$\left.+\frac{(2)^3\,(113)}{t}\right)p0+\left((2)^2\,(3)\,(7)^{10}\,t^5+(2)\,(5)\,(7)^8\,(43)\,t^4+(2)\,(3)\,(7)^7\,(109)\,t^3\right.$$

$$\left.+(7)^5\,(433)\,t^2-(2)\,(5)\,(7)^4\,(229)\,t-(3)^4\,(7)\,(67)+\frac{(2)\,(3)\,(43)\,(113)}{t}\right.$$

$$\left.-\frac{(113)}{t^2}\right)p1-(7)^{11}\,t^6-(2)^2\,(7)^9\,(31)\,t^5-(2)\,(5)\,(7)^7\,(11)\,(31)\,t^4$$

$$-(2)\,(3)\,(7)^6\,(17)\,(43)\,t^3-(3)\,(5)\,(7)^4\,(131)\,t^2+(2)\,(7)^2\,(13)\,(41)\,(101)\,t$$

$$+(19)\,(23)\,(89)-\frac{(2)\,(3)\,(43)\,(113)}{t}+\frac{(113)}{t^2}$$

```
> # GROUP 4, IDENTITY 1: U_B7(q*p0)
  m := 5 :
```

```
S7_4_1 := series(B7q * q * p0q, q, 7 * T) :
target7_4_1 := add(coeff(S7_4_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_4_1 := expand(findlincombo(target7_4_1, basis7, symbols7, q, 0)) :
collect(R7_4_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((2)^3\ (7)^7\ t^4 + (2)^4\ (5)\ (7)^5\ t^3 + (2)^3\ (3)^3\ (7)^3\ t^2 + (7)\ (79)\ t\right) p0 + \left((7)^{10}\ t^5 + (7)^8\ (19)\ t^4 + (2)\ (3)^2\ (7)^7\ t^3 + (3)\ (7)^4\ (109)\ t^2 + (2)\ (7)^3\ (17)\ t + (2)^2\right) p1 + (7)^9\ t^5 + (7)^7\ (11)\ t^4 + (2)\ (7)^5\ (19)\ t^3 + (7)^3\ (31)\ t^2 + (2)\ (3)\ (7)\ t \tag{22}$$

```
> # GROUP 4, IDENTITY 2: U_B7(q*p0*t^(-1))
  m := 5 :
  S7_4_2 := series(B7q * q * p0q / tq, q, 7 * T) :
  target7_4_2 := add(coeff(S7_4_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R7_4_2 := expand(findlincombo(target7_4_2, basis7, symbols7, q, 0)) :
  collect(R7_4_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(7)^3\ t^2 - (2)^3\ (7)\ t\right) p0 + \left((7)^2\ t + (7)\right) p1 + 1 - (7)^2\ t \tag{23}$$

```
> # GROUP 4, IDENTITY 3: U_B7(q*p0*t^(-2))
  m := 5 :
  S7_4_3 := series(B7q * q * p0q / tq^2, q, 7 * T) :
  target7_4_3 := add(coeff(S7_4_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R7_4_3 := expand(findlincombo(target7_4_3, basis7, symbols7, q, 0)) :
  collect(R7_4_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(7)^5\ t^3 - (2)^3\ (7)^3\ t^2\right) p0 + \left((7)^4\ t^2 + (7)^3\ t\right) p1 - (7)^4\ t^2 - (7)^3\ t - (2)^2\ (3) \tag{24}$$

```
> # GROUP 4, IDENTITY 4: U_B7(q*p0*t^(-3))
  m := 5 :
  S7_4_4 := series(B7q * q * p0q / tq^3, q, 7 * T) :
```

```
target7_4_4 := add(coeff(S7_4_4, q, 7*n)*q^n, n=0..T-1) + O(q^T):
basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)]:
symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)]:
R7_4_4 := expand(findlincombo(target7_4_4, basis7, symbols7, q, 0)):
collect(R7_4_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(1 + (2)\,(3)\,(7)^7\,t^4 + (2)\,(7)^5\,(31)\,t^3 + (3)^3\,(7)^4\,t^2 + (2)\,(5)\,(7)^2\,t\right) p0 + \left(\right. \tag{25}$$
$$\left. -(2)\,(3)\,(7)^6\,t^3 - (2)^3\,(7)^5\,t^2 - (3)\,(7)^4\,t - (2)^2\,(7)\right) p1 + (2)\,(7)^6\,t^3$$
$$+ (2)^3\,(3)\,(7)^4\,t^2 + (7)^2\,(83)\,t + (2)^2\,(3)^3$$

```
> # GROUP 4, IDENTITY 5: U_B7(q*p0*t^(-4))
m := 5:
S7_4_5 := series(B7q*q*p0q/tq^4, q, 7*T):
target7_4_5 := add(coeff(S7_4_5, q, 7*n)*q^n, n=0..T-1) + O(q^T):
basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)]:
symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)]:
R7_4_5 := expand(findlincombo(target7_4_5, basis7, symbols7, q, 0)):
collect(R7_4_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^9\,t^5 - (2)\,(5)^2\,(7)^7\,t^4 - (2)^2\,(3)\,(5)\,(7)^6\,t^3 - (2)\,(3)^4\,(7)^4\,t^2 - (2)^2\,(3)\,(5)\,(7)^2\,t\right. \tag{26}$$
$$\left. - (2)\,(3)\right) p0 + \left((7)^8\,t^4 + (7)^8\,t^3 + (2)\,(3)^3\,(7)^5\,t^2 + (2)\,(3)^2\,(7)^4\,t\right.$$
$$\left. + (2)^3\,(3)\,(7)\right) p1 - (7)^8\,t^4 - (5)^2\,(7)^6\,t^3 - (2)\,(3)\,(7)^4\,(31)\,t^2 - (2)\,(3)\,(7)^2\,(83)\,t$$
$$- (2)^3\,(3)\,(5)\,(7)$$

```
> # GROUP 4, IDENTITY 6: U_B7(q*p0*t^(-5))
m := 6:
S7_4_6 := series(B7q*q*p0q/tq^5, q, 7*T):
target7_4_6 := add(coeff(S7_4_6, q, 7*n)*q^n, n=0..T-1) + O(q^T):
basis7 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)]:
symbols7 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)]:
R7_4_6 := expand(findlincombo(target7_4_6, basis7, symbols7, q, 0)):
collect(R7_4_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left(-(7)^{11}\,t^6 - (2)^3\,(7)^9\,t^5 + (3)\,(7)^9\,t^4 + (2)\,(3)\,(5)\,(7)^7\,t^3 + (3)^4\,(7)^5\,t^2\right. \tag{27}$$

$$+ (2)\,(3)\,(5)\,(7)^3\,t + (3)\,(7))\,p0 + ((7)^{10}\,t^5 + (7)^9\,t^4 - (3)\,(7)^8\,t^3 - (3)^3\,(7)^6\,t^2$$
$$- (3)^2\,(7)^5\,t - (2)^2\,(3)\,(7)^2)\,p1 - (7)^{10}\,t^5 - (7)^9\,t^4 + (3)^2\,(7)^7\,t^3 + (3)\,(7)^5\,(31)\,t^2$$
$$+ (3)\,(7)^3\,(83)\,t + (2)^2\,(7)\,(11)\,(19)$$

> # GROUP 4, IDENTITY 7: U_B7(q*p0*t^(-6))

```
m := 7 :
S7_4_7 := series(B7q * q * p0q / tq^6, q, 7 * T) :
target7_4_7 := add(coeff(S7_4_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_4_7 := expand(findlincombo(target7_4_7, basis7, symbols7, q, 0)) :
collect(R7_4_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$
$$\#\ of\ terms\ , 45$$

$$(-(7)^{13}\,t^7 - (2)^3\,(7)^{11}\,t^6 + (2)\,(7)^9\,t^4 + (2)^2\,(5)\,(7)^7\,t^3 + (2)\,(3)^3\,(7)^5\,t^2 \qquad (28)$$
$$+ (2)^2\,(5)\,(7)^3\,t + (2)\,(7))\,p0 + ((7)^{12}\,t^6 + (7)^{11}\,t^5 - (2)\,(7)^8\,t^3 - (2)\,(3)^2\,(7)^6\,t^2$$
$$- (2)\,(3)\,(7)^5\,t - (2)^3\,(7)^2)\,p1 - (7)^{12}\,t^6 - (7)^{11}\,t^5 + (2)\,(3)\,(7)^7\,t^3$$
$$+ (2)\,(7)^5\,(31)\,t^2 + (2)\,(7)^3\,(83)\,t - (2)^2\,(7)^2\,(11)\,(17)$$

> # GROUP 5, IDENTITY 1: U_B7(q*p1*t^(-1))

```
m := 5 :
S7_5_1 := series(B7q * q * p1q / tq, q, 7 * T) :
target7_5_1 := add(coeff(S7_5_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_5_1 := expand(findlincombo(target7_5_1, basis7, symbols7, q, 0)) :
collect(R7_5_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$
$$\#\ of\ terms\ , 39$$

$$((2)^3\,(7)^8\,t^4 + (2)^4\,(5)\,(7)^6\,t^3 + (2)^3\,(3)^3\,(7)^4\,t^2 + (2)^3\,(3)\,(7)\,(23)\,t)\,p0 + ((7)^{11}\,t^5 \qquad (29)$$
$$+ (7)^9\,(19)\,t^4 + (2)\,(3)^2\,(7)^8\,t^3 + (3)\,(7)^5\,(109)\,t^2 + (2)\,(7)^4\,(17)\,t + (29))\,p1 - 1$$
$$+ (7)^{10}\,t^5 + (7)^8\,(11)\,t^4 + (2)\,(7)^6\,(19)\,t^3 + (7)^4\,(31)\,t^2 + (7)\,(41)\,t$$

> # GROUP 5, IDENTITY 2: U_B7(q*p1*t^(-2))

```
m := 5 :
S7_5_2 := series(B7q * q * p1q / tq^2, q, 7 * T) :
target7_5_2 := add(coeff(S7_5_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
```

```
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_5_2 := expand(findlincombo(target7_5_2, basis7, symbols7, q, 0)) :
collect(R7_5_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^3\,t^2-(2)^3\,(7)^2\,t\right)p0+\left((2)^3\,(7)^2\,t+(7)^2\right)p1-(7)^3\,t^2-(2)^3\,(7)^2\,t+(11) \quad (30)$$

```
> # GROUP 5, IDENTITY 3: U_B7(q*p1*t^(-3))
  m := 5 :
  S7_5_3 := series(B7q * q * p1q / tq^3, q, 7 * T) :
  target7_5_3 := add(coeff(S7_5_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R7_5_3 := expand(findlincombo(target7_5_3, basis7, symbols7, q, 0)) :
  collect(R7_5_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^5\,t^3-(2)^3\,(7)^4\,t^2\right)p0+\left((2)^3\,(7)^4\,t^2+(7)^4\,t\right)p1-(7)^5\,t^3-(2)^3\,(7)^4\,t^2 \quad (31)$$
$$-(7)^4\,t-(2)^5\,(3)$$

```
> # GROUP 5, IDENTITY 4: U_B7(q*p1*t^(-4))
  m := 5 :
  S7_5_4 := series(B7q * q * p1q / tq^4, q, 7 * T) :
  target7_5_4 := add(coeff(S7_5_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R7_5_4 := expand(findlincombo(target7_5_4, basis7, symbols7, q, 0)) :
  collect(R7_5_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^4\,(3)\,(7)^7\,t^4+(2)^3\,(3)^2\,(7)^6\,t^3+(2)^3\,(3)^3\,(7)^4\,t^2+(2)^4\,(5)\,(7)^2\,t+(2)^3\right)p0+\left(\right. \quad (32)$$
$$\left.-(2)^4\,(3)\,(7)^6\,t^3-(5)\,(7)^5\,(13)\,t^2-(2)^3\,(3)\,(7)^4\,t-(2)^5\,(7)\right)p1-(7)^7\,t^4$$
$$+(2)^4\,(7)^6\,t^3+(7)^4\,(199)\,t^2+(2)^3\,(7)^2\,(83)\,t+(2)^2\,(5)\,(37)$$

```
> # GROUP 5, IDENTITY 5: U_B7(q*p1*t^(-5))
  m := 5 :
  S7_5_5 := series(B7q * q * p1q / tq^5, q, 7 * T) :
  target7_5_5 := add(coeff(S7_5_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
```

```
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_5_5 := expand(findlincombo(target7_5_5, basis7, symbols7, q, 0)) :
collect(R7_5_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\begin{aligned}&\left(-(2)^3\,(7)^9\,t^5-(2)^6\,(7)^8\,t^4-(2)^4\,(5)\,(7)^7\,t^3-(2)^3\,(3)^3\,(7)^5\,t^2-(2)^4\,(5)\,(7)^3\,t\right.\\&\left.-(2)^3\,(7)\right)p0+\left((2)^3\,(7)^8\,t^4+(3)^2\,(7)^8\,t^3+(2)^3\,(3)^2\,(7)^6\,t^2+(2)^3\,(3)\,(7)^5\,t\right.\\&\left.+(2)^5\,(7)^2\right)p1-(7)^9\,t^5-(2)^3\,(7)^8\,t^4-(7)^7\,(31)\,t^3-(2)^3\,(7)^5\,(31)\,t^2\\&-(2)^3\,(7)^3\,(83)\,t-(2)^2\,(5)\,(7)\,(37)\end{aligned}\qquad\textbf{(33)}$$

```
> # GROUP 5, IDENTITY 6: U_B7(q*p1*t^(-6))
m := 6 :
S7_5_6 := series(B7q * q * p1q / tq^6, q, 7 * T) :
target7_5_6 := add(coeff(S7_5_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_5_6 := expand(findlincombo(target7_5_6, basis7, symbols7, q, 0)) :
collect(R7_5_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&\left(-(2)^3\,(7)^{11}\,t^6-(2)^3\,(7)^{10}\,t^5+(2)^3\,(5)\,(7)^9\,t^4+(2)^4\,(5)^2\,(7)^7\,t^3+(2)^3\,(3)^3\,(5)\,(7)^5\,t^2\right.\\&\left.+(2)^4\,(5)^2\,(7)^3\,t+(2)^3\,(5)\,(7)\right)p0+\left((2)^3\,(7)^{10}\,t^5+(7)^{10}\,t^4-(2)^3\,(5)\,(7)^8\,t^3\right.\\&\left.-(2)^3\,(3)^2\,(5)\,(7)^6\,t^2-(2)^3\,(3)\,(5)\,(7)^5\,t-(2)^5\,(5)\,(7)^2\right)p1-(7)^{11}\,t^6\\&-(2)^3\,(7)^{10}\,t^5-(7)^{10}\,t^4+(2)^3\,(3)\,(5)\,(7)^7\,t^3+(2)^3\,(5)\,(7)^5\,(31)\,t^2\\&+(2)^3\,(5)\,(7)^3\,(83)\,t+(2)^4\,(5)\,(7)\,(59)\end{aligned}\qquad\textbf{(34)}$$

```
> # GROUP 5, IDENTITY 7: U_B7(q*p1*t^(-7))
m := 8 :
S7_5_7 := series(B7q * q * p1q / tq^7, q, 7 * T) :
target7_5_7 := series(tq * (add(coeff(S7_5_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T)), q, T) :
basis7 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R7_5_7 := expand(findlincombo(target7_5_7, basis7, symbols7, q, 0) / t) :
collect(R7_5_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

$$\left(-(2)^3\,(7)^{13}\,t^7-(2)^3\,(7)^{12}\,t^6-(2)^3\,(7)^9\,(17)\,t^4-(2)^4\,(5)\,(7)^7\,(17)\,t^3\right.\qquad\textbf{(35)}$$

$$-(2)^3\,(3)^3\,(7)^5\,(17)\,t^2-(2)^4\,(5)\,(7)^3\,(17)\,t-(2)^3\,(7)\,(17))\,p0+\Big((2)^3\,(7)^{12}\,t^6$$

$$+(2)\,(3)\,(7)^{11}\,t^5-(7)^9\,(19)\,t^4+(2)\,(7)^7\,(409)\,t^3+(3)\,(7)^6\,(389)\,t^2$$

$$+(2)\,(7)^4\,(1399)\,t+(7)^4\,(11)+\frac{1}{t}\Big)\,p1+(2)\,(3)\,(7)^{13}\,t^7+(2)^2\,(5)\,(7)^{12}\,t^6$$

$$+(2)^4\,(7)^{10}\,(17)\,t^5+(3)\,(7)^8\,(631)\,t^4+(3)\,(5)\,(7)^7\,(29)\,t^3-(7)^5\,(2953)\,t^2$$

$$-(2)^2\,(3)^2\,(5)\,(7)^3\,(59)\,t-(7)\,(151)\,(179)$$

```
> # GROUP 6, IDENTITY 1: U_B7(q)
  m := 1 :
  S7_6_1 := series(B7q * q, q, 7 * T) :
  target7_6_1 := add(coeff(S7_6_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_1 := expand(findlincombo(target7_6_1, basis7, symbols7, q, 0)) :
  collect(R7_6_1, [t], recursive, ifactor);
```

$$nx = , 2$$

$$\#\ of\ terms\ , 23$$

$$1 \tag{36}$$

```
> # GROUP 6, IDENTITY 2: U_B7(q*t^(-1))
  m := 1 :
  S7_6_2 := series(B7q * q / tq, q, 7 * T) :
  target7_6_2 := add(coeff(S7_6_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_2 := expand(findlincombo(target7_6_2, basis7, symbols7, q, 0)) :
  collect(R7_6_2, [t], recursive, ifactor);
```

$$nx = , 2$$

$$\#\ of\ terms\ , 23$$

$$-(7)\,t-(2)^2 \tag{37}$$

```
> # GROUP 6, IDENTITY 3: U_B7(q*t^(-2))
  m := 2 :
  S7_6_3 := series(B7q * q / tq^2, q, 7 * T) :
  target7_6_3 := add(coeff(S7_6_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_3 := expand(findlincombo(target7_6_3, basis7, symbols7, q, 0)) :
  collect(R7_6_3, [t], recursive, ifactor);
```

$$nx = , 3$$

$$\#\ of\ terms\ ,\ 24$$

$$-(7)^3\ t^2 + (2)^2\ (5) \qquad (38)$$

```
> # GROUP 6, IDENTITY 4: U_B7(q*t^(-3))
  m := 3 :
  S7_6_4 := series(B7q * q / tq^3, q, 7 * T) :
  target7_6_4 := add(coeff(S7_6_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_4 := expand(findlincombo(target7_6_4, basis7, symbols7, q, 0)) :
  collect(R7_6_4, [t], recursive, ifactor);
```

$$nx\ =\ ,\ 4$$

$$\#\ of\ terms\ ,\ 25$$

$$-(7)^5\ t^3 - (2)^3\ (11) \qquad (39)$$

```
> # GROUP 6, IDENTITY 5: U_B7(q*t^(-4))
  m := 4 :
  S7_6_5 := series(B7q * q / tq^4, q, 7 * T) :
  target7_6_5 := add(coeff(S7_6_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_5 := expand(findlincombo(target7_6_5, basis7, symbols7, q, 0)) :
  collect(R7_6_5, [t], recursive, ifactor);
```

$$nx\ =\ ,\ 5$$

$$\#\ of\ terms\ ,\ 26$$

$$-(7)^7\ t^4 + (2)^2\ (5)\ (13) \qquad (40)$$

```
> # GROUP 6, IDENTITY 6: U_B7(q*t^(-5))
  m := 5 :
  S7_6_6 := series(B7q * q / tq^5, q, 7 * T) :
  target7_6_6 := add(coeff(S7_6_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
  symbols7 := [seq(t^j, j = 0 ..m)] :
  R7_6_6 := expand(findlincombo(target7_6_6, basis7, symbols7, q, 0)) :
  collect(R7_6_6, [t], recursive, ifactor);
```

$$nx\ =\ ,\ 6$$

$$\#\ of\ terms\ ,\ 27$$

$$-(7)^9\ t^5 + (2)^2\ (7)\ (17) \qquad (41)$$

```
> # GROUP 6, IDENTITY 7: U_B7(q*t^(-6))
  m := 6 :
  S7_6_7 := series(B7q * q / tq^6, q, 7 * T) :
  target7_6_7 := add(coeff(S7_6_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
```

```
basis7 := [seq(series(tq^j, q, T), j = 0 ..m)] :
symbols7 := [seq(t^j, j = 0 ..m)] :
R7_6_7 := expand(findlincombo(target7_6_7, basis7, symbols7, q, 0)) :
collect(R7_6_7, [t], recursive, ifactor);
```

$$nx = , 7$$

$$\#\ of\ terms\ , 28$$

$$-(7)^{11}\, t^6 - (2)^3\, (7)\, (13)\, (23) \qquad (42)$$

>